

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	11 July 2026
Team ID	
Project Name	Customer Care Registry
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per Table-1 & Table-2.

Customer Care Registry — MERN Stack Architecture

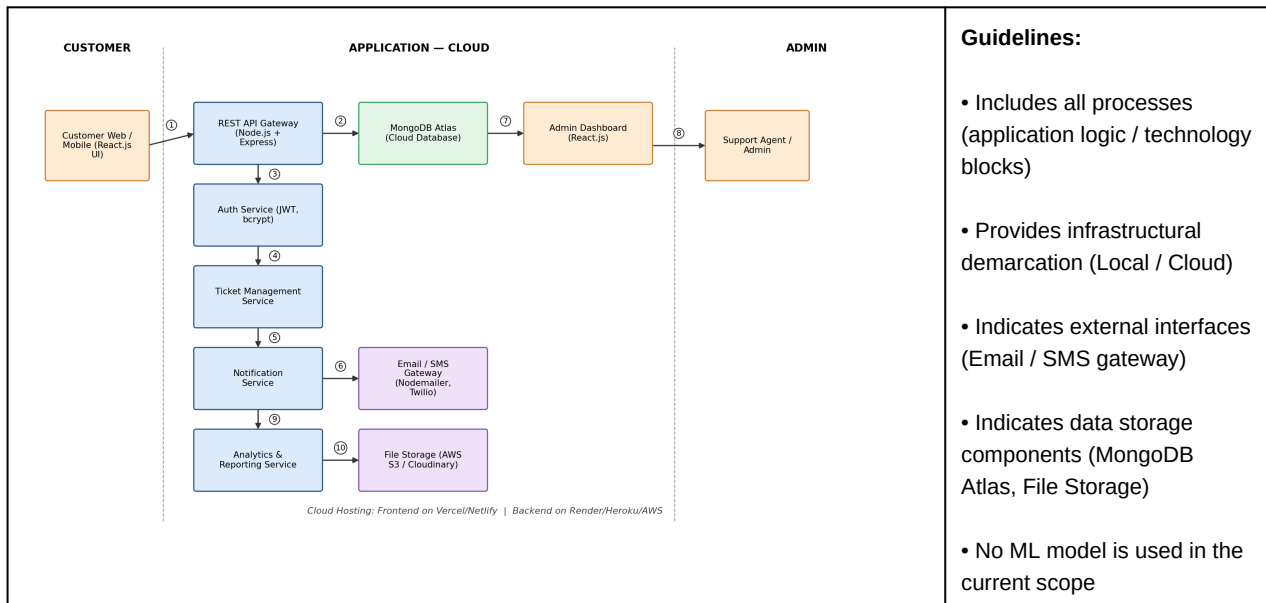


Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	How the customer interacts with the application — web portal to raise, track, and view complaints/tickets.	HTML5, CSS3, JavaScript, React.js, Bootstrap/Tailwind CSS
2.	Application Logic-1	User registration, login and role-based authentication/authorization.	Node.js, Express.js, JWT, bcrypt
3.	Application Logic-2	Ticket/complaint creation, categorization, assignment and status management.	Node.js, Express.js REST APIs
4.	Application Logic-3	Automated notifications to customers on ticket status changes.	Node.js (Nodemailer), Twilio API
5.	Database	Stores users, tickets, feedback and interaction history.	MongoDB, Mongoose ODM
6.	Cloud Database	Managed database service on cloud.	MongoDB Atlas
7.	File Storage	Storing attachments/screenshots uploaded with a complaint.	AWS S3 / Cloudinary
8.	External API-1	Sending email notifications and confirmations.	SMTP / Nodemailer / SendGrid API
9.	External API-2	Sending SMS alerts to customers on urgent updates.	Twilio API
10.	Machine Learning Model	Auto-categorizing / prioritizing complaints (future scope).	Keyword-based NLP classification (Node NLP)

S.No	Component	Description	Technology
11.	Infrastructure (Server / Cloud)	Application deployment. Local Server Configuration: Node.js v18, npm, local MongoDB instance. Cloud Server Configuration: Render / Heroku dyno + MongoDB Atlas cluster.	Local, Render/Heroku, Vercel/Netlify

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source frameworks used to build the application end-to-end.	React.js, Node.js, Express.js, Mongoose (MERN Stack)
2.	Security Implementations	Access controls and safeguards implemented across the application.	bcrypt password hashing, JWT auth, HTTPS/SSL, express-validator, CORS policy
3.	Scalable Architecture	3-tier architecture (Presentation – React, Application – Node/Express, Data – MongoDB) allows each layer to scale independently.	Stateless REST APIs, containerizable with Docker for horizontal scaling
4.	Availability	Cloud hosting with managed database replication ensures high uptime.	MongoDB Atlas replica sets, cloud auto-restart & monitoring
5.	Performance	Optimized queries, indexing and pagination for fast response even with large ticket volumes.	MongoDB indexing, Redis caching (optional), CDN for static assets

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>